



Assessment Template

Computing

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Assessment:

Item	Description
Module Title:	Graph & AI
Module Code:	B9AI101
Module Lecturer (s):	Terri Hoare
Programme/Cohort:	MSc Artificial Intelligence
Method of Assessment:	Artefact and Report
Percentage (%) Weighting:	50%
MIMLOs being assessed:	MIMLO1; MIMLO3; MIMLO05
Assessment Number:	1
Individual/Group:	Individual
Issue Date:	Week 1
Submission Date:	Sunday 2 November 2025
Feedback Date:	Week 10
Feedback Strategy:	Moodle

Assignment Task**[100 Marks]****Task 1 - Research topic in graph learning**

Select and critique a recent journal paper on a topic in Graph Learning in AI. There are two main purposes in doing this assignment:

- (i) to develop your ability to critically read a technical topic and summarize your view of it.
- (ii) to build on your awareness of state of the art in Graph Learning in AI research

The research paper should be summarized under the following seven headings:

Context (topic, motivation; arising from literature review)

(Where is this paper coming from (what is the research topic)?; what are the main issues/themes in this topic?; what prior academic work is relevant to this research topic and have the authors adequately referenced it? where is the envelope or boundary between what is known and what is not known in this area?)

Gap (point of departure, “what next?”)

(Does the paper identify a hole in the context/literature which it attempts to fill?; does the paper extend the envelope to cover something previously unknown?)

Question (research aim)

(States the problem / question that this paper attempts to solve / answer; have the authors rephrased the themes as questions to which an answer can be found (with reasonable certainty) using their chosen research instrument)

How did they do the work? (method(ology), research approach)

(What process did they use to solve the problem, that is, what was the academically valid way that they got from question to answer; is the work technically sound; was it mathematical, empirical (survey or experimental) or some other method - if experimental, did they explain how the experimental design tests the research hypotheses, and is the work reproducible (enough detail is given so someone else could carry out the same work)

What did they find? (results)

(Presents the “data”/argumentation the authors collected/created; what is the answer (analysis, discussion); what answer is suggested by the data; what is their meaning, in both academic and practical contexts; can they be sure: how do they validate their results; how would another person (reviewer/examiner) arrive at an objective assessment of this work; have they established or reinforced any principles, or can they make any generalisations; does the data allow them to give underlying causes; can they only observe statistical regularities (correlation); does the paper achieve its stated objective - does it answer the research question)

Significance (“so what?”, conclusion, value add)

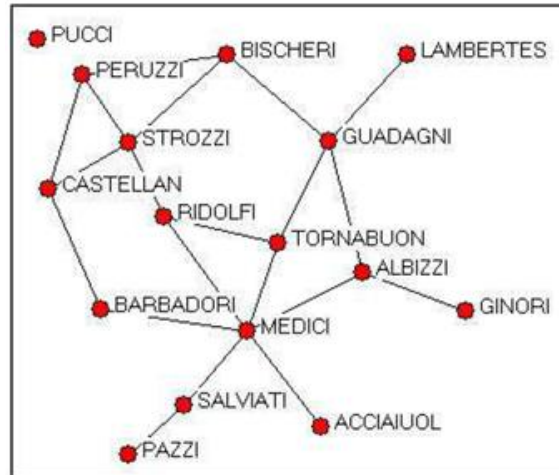
(why is this paper important; what does it contribute; how is the contribution original or innovative; what value does the answer have for business, academics; since the paper was published, what has been its impact on academia and/or practice - number of citations, quality of journal, etc).

The critique should be at most two pages of text containing your assessment and conclusion, no smaller than 10-point font. The cover page does not count towards the page limit.

(30 marks)

Task 2 - Graph Algorithm Application and Knowledge Graph Tool Familiarization

Padgett's data on marriage alliances among leading Florentine families in the latter Renaissance is given in the Figure below. Each node is a family, and each edge denotes a connection by marriage.



For the Medici Family, hand calculate degree centrality, closeness centrality, and betweenness centrality. Show all calculations.

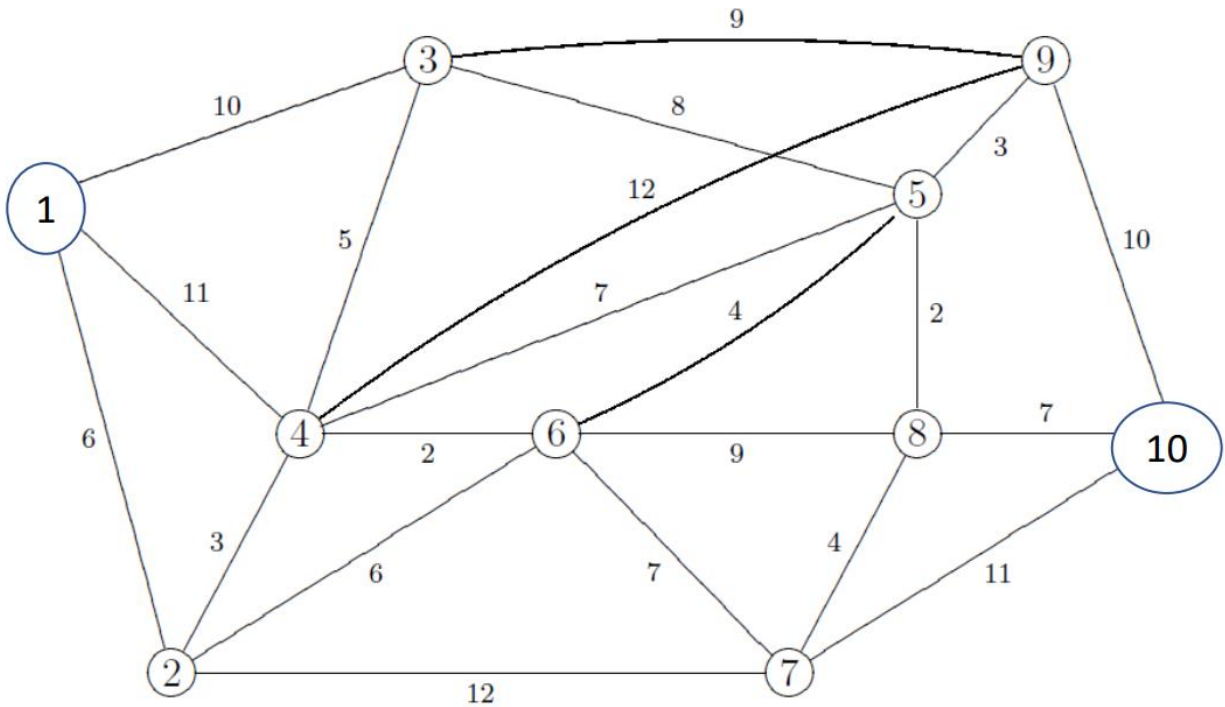
Load the data as a Neo4j Knowledge Graph. For each node, compute and interpret the following measures of centrality: Degree Centrality; Closeness Centrality; Betweenness Centrality; Eigenvector Centrality; PageRank. Use Neo4j Bloom to visualize your insights.

(50 marks)

Task 3 - Graph Algorithm Application and Python Library Familiarisation

Consider the street network depicted below. Each edge represents a street, each node represents a street junction, and each edge weight represents street length in hundreds of meters. Each street is a 2-way street. The town council is constructing a new town center at node 10, and wants to transport construction material available at node 1 to node 10. This will take several trips, so it wants to find the shortest path from node 1 to node 10.

Apply Dijkstra's Shortest Path Algorithm to hand calculate the shortest path from node 1 to node 10. Cross-check and visualize your result using Python Network-X.



(20 marks)

Submission

Include a cover page on your report and cite all references. Two files should be loaded to Moodle on or before Sunday 2 November 2025 (23:59).

1. A SINGLE pdf file named CA01_Surname_First Name_Student-ID
2. A zipped file including your code for tasks 2 and 3.

Overall marks will be awarded according to the following criteria:

001: Identification/analysis under specified headings of a current research topic in GL.	30%
002: Centrality Algorithm calculation and interpretation.	30%
003: Path Finding Algorithm calculation and interpretation.	10%
004: Application of Tools - Neo4j and Python Network-X	30%

Resources

Software or tools required: Neo4j GDS and Python Network-X

1	2	3	4	5
NO AI	AI-ASSISTED IDEA GENERATION AND STRUCTURING	AI-ASSISTED EDITING	AI TASK COMPLETION, HUMAN EVALUATION	FULL AI
The assessment is completed entirely without AI assistance. This level ensures that students rely solely on their knowledge, understanding, and skills. AI must not be used at any point during the assessment.	AI can be used in the assessment for brainstorming, creating structures, and generating ideas for improving work. No AI content is allowed in the final submission.	AI can be used to make improvements to the clarity or quality of student created work to improve the final output, but no new content can be created using AI. AI can be used, but your original work with no AI content must be provided in an appendix.	AI is used to complete certain elements of the task, with students providing discussion or commentary on the AI-generated content. This level requires critical engagement with AI generated content and evaluating its output. You will use AI to complete specified tasks in your assessment. Any AI created content must be cited.	AI should be used as a 'co-pilot' in order to meet the requirements of the assessment, allowing for a collaborative approach with AI and enhancing creativity. You may use AI throughout your assessment to support your own work and do not have to specify which content is AI generated

		Evaluative Criteria/Performance Level Descriptors (what does performance at this level look like?)			
Scale	Grade Band	Criterion 1 - "001"	Criterion 2 - "002"	Criterion 3 - "003"	Criterion 4 - "004"
Exceptional	90-100%	Exceeding module requirements	Exceeding module requirements	Exceeding module requirements	Exceeding module requirements
Outstanding	80-89%	Evidence of non-core advanced knowledge	Evidence of non-core advanced knowledge and / or interpretation of findings	Evidence of non-core advanced knowledge and / or interpretation of findings	Evidence of non-core advanced knowledge on use of tools
Excellent	70-79%	Very good both technically and a well-presented critical interpretation	Accurate with a well-presented critical interpretation	Accurate with a well-presented critical interpretation	Very good and a sound use of tools
Very Good	60-69%	Evidence of some critical interpretation beyond that covered in class	Evidence of some critical interpretation beyond that covered in class	Evidence of some critical interpretation beyond that covered in class	Evidence of some knowledge of tool use beyond that covered in class
Good	50-59%	Evidence of understanding of core concepts covered under headings	Evidence of understanding of core concepts covered in class	Evidence of understanding of core concepts covered in class	Evidence of understanding of core concepts of tool use
Sufficient	40-49%	Submission met basic requirements	Submission met basic requirements	Submission met basic requirements	Submission met basic requirements
Somewhat/ Unfulfilled	30-39%	Submission not completed in full	Submission not completed in full	Submission not completed in full	Submission not completed in full
Unfulfilled	0-29%	Submission not completed in full and poorly presented	Submission not completed in full and poorly presented	Submission not completed in full and poorly presented	Submission not completed in full and poorly presented
MIMLO Alignment (see Module Guide for Minimum Intended Learning Outcomes for Module)					
MIMLO criterion is aligned:		MIMLO03	MIMLO01	MIMLO01	MIMLO05

General Requirements for Students:

1. All assignments must be submitted no later than the stated deadline (date and time).
2. Assignments submitted after the latest deadline specified (including any approved extension deadline) are considered late and penalised according to the Quality Assurance Handbook (QAH) Part B Section 5.2.2.6 as follows:
 - a. A penalty of 2 marks will be applied per day or part thereof (including weekends and public holidays) for an ongoing failure to submit beyond the submission deadline.
 - b. An examiner has the right to refuse to mark the assignment if the submission instructions have not been observed.
 - c. Where a late assessment is submitted within 14 days of the deadline, and is of a passing standard, the late penalty is capped (such that the minimum grade that can be awarded is 40% for the late submission).
 - d. Where a late assessment is submitted more than 14 days after the deadline, it will receive 0%. The lecturer may, at their discretion, review the submission for feedback.
 - e. Where the assessment is undertaken in a group, the piece of work should be submitted in its complete entirety, and any penalty for late submission incurred applies to all group members.
3. Extensions to assignment submission deadlines will not be granted, other than in exceptional circumstances. To apply for an extension please go to <https://students.dbs.ie/dashboard/SCCM> and open a ticket.
4. All relevant provisions of the Assessment Regulations must be complied with (see QAH B.5).
 - a. Students are required to refer to the assessment regulations in their Programme Handbook, and on the Student Website.
 - b. Dublin Business School penalises students who engage in academic impropriety (i.e. plagiarism, collusion and/or copying, ghost writing/ essay mills, improper use of Generative Artificial Intelligence software).
 - i. Refer to the College's Generative AI Guidelines HERE for further information.
 - c. Guides on referencing are available on the Library website: <https://libguides.dbs.ie/referencing>
 - d. Text-matching analysis software is integrated in Moodle to generate a report regarding the degree of text-matching in a submission.
5. Students are required to retain a copy of each assignment submitted, until the issuing of a transcript indicating the mark awarded and the closure of the Appeal period (2 weeks following the release of final results).
 - a. Results can only be appealed following the release of final results, and the Appeal form must be submitted to the Exams Office within the Appeal period.
 - b. An appeal must be based on valid grounds (see the Appeals Policy QAH B.3.5), dissatisfaction with a grade is not sufficient grounds for an appeal.
 - c. Assignments must be appropriately packaged and presented.

- d. All assignments should be submitted to your subject/course page on Moodle by the deadline date.
 - e. Where a submission involves digital media (i.e formats other than Word, Powerpoint or PDF), it is the submitting students' responsibility to ensure the media is appropriately labelled, fully working and they must retain a copy.
 - f. Components of an assessment which are not included in the final submission cannot normally be subsequently accepted for grading. It is the student's responsibility to ensure their file is uploaded correctly.
 - g. Include an electronic **cover sheet** with the following details to the front of the assignment (see below)
6. Assignments that *breach* the word count requirements will be penalised. *There is a 10% discretion, either way, applicable in terms of word count.*
7. When you submit your assignment you will be asked to click on a button which will declare the following:

By ticking this box I am confirming that this assignment/exam is all my own work. Any sources used have been referenced.

I have read the College rules regarding plagiarism in the QAH Part B Section 3 and understand that penalties will be applied accordingly if work is found not to be my own. All work uploaded is submitted via Ouriginal, whereby a text-matching report will show any similarities with other texts.